A

Mini Project Report

on

Securio

Submitted in partial fulfillment of the requirements for the degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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UNIVERSITY OF MUMBAI

Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on Securio has been submitted by Pratiksha Jadhav(24107107),Shravani Bhilare (24107112), Srushti Balsaraf(24107096) and Sakshi Kale(24107071) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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1.

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Place: A. P. Shah Institute of Technology, Thane

Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide Prof.Veena Gawde Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1: Introduction

In today's digital era, ensuring security and efficient management in residential societies has become increasingly important. However, many apartment complexes still rely on manual methods such as maintaining visitor registers, calling residents for approvals, and handling complaints and maintenance records offline. These traditional systems are not only time-consuming but also lack transparency, real-time communication, and proper data management, leading to security risks and operational inefficiencies.

The project **Securio – Apartment Security & Management App** aims to provide a centralized and digital solution to overcome these challenges. The application focuses on improving apartment security through a structured visitor management system while enabling seamless communication between guards, residents, and administrators. By digitizing these processes, the system reduces manual effort and enhances overall efficiency.

The app introduces a role-based approach where Admin, Guard, and Owner have specific functionalities. Guards can register visitors and notify owners instantly, owners can approve or reject entries in real-time, and admins can manage complaints, maintenance, and user data effectively. This ensures smooth coordination and faster decision-making within the society.

The major contribution of this project is the integration of key apartment management features into a single platform, including real-time visitor approval, complaint handling with status tracking, maintenance monitoring, and house helper management. This system improves security, ensures transparency, and provides a modern and user-friendly solution for apartment management.

1.1 Purpose:

The purpose of this document is to present a detailed description of the **Securio – Apartment Security & Management App**, including its design, features, functionality, and implementation. It serves as a comprehensive guide to understand how the system works, the problems it solves, and the technologies used in its development.

This document is intended for multiple stakeholders, including:

- **Project Evaluators and Faculty:** To assess the design, implementation, and outcomes of the project
- **Developers:** To understand the system architecture, modules, and workflow for further development or maintenance
- **Apartment Administrators:** To understand how the application can be used for efficient society management
- **Residents (Owners):** To gain insight into how the system improves security, communication, and convenience
- **Security Personnel (Guards):** To understand the usage of the visitor management system

Overall, this document acts as a reference for understanding, evaluating, and potentially enhancing the application.

1.2 Problem Statement:

In many residential apartment societies, daily operations such as visitor management, complaint handling, maintenance tracking, and communication are still managed using manual and outdated systems. These methods involve physical registers, phone calls, and paper-based records, which are inefficient, time-consuming, and prone to human errors.

The lack of a proper digital system leads to several issues, including poor visitor verification, delays in communication between guards and residents, lack of transparency in complaint resolution, and difficulty in tracking maintenance records. Additionally, there is no structured mechanism to manage and monitor house helpers, which raises security concerns. Due to these limitations, there is a need for a centralized and automated solution that ensures secure data handling, real-time communication, and efficient management of apartment activities.

1.3 Objectives:

1. To prevent unauthorized access in residential buildings.
2. To enable real-time resident approval for visitor entry.
3. To digitize and automate the visitor entry process.
4. To maintain accurate visitor entry logs for future reference

1.4 Scope

The scope of this project is limited to developing a mobile-based solution for apartment societies. The application covers:

1. Can be applied in residential societies, apartments, and gated communities for digital visitor management.

2. Can be useful to security staff for registering and tracking visitor entries efficiently.

3. Can assist residents in approving or rejecting visitors through a mobile application.

4. Can be extended in future to include features like emergency alerts and advanced security monitoring.

Chapter 2: Proposed System

The proposed system is a smart apartment management application that automates all major operations of a residential society. It replaces manual processes with a digital system that ensures better security, faster communication, and efficient data management.The system uses Firebase for authentication and real-time database management, while Cloudinary is used for efficient image handling.

2.1 Features and Functionality

- **Visitor Management:**

  Guards can add visitor details, and owners can approve or reject requests instantly.

- **Real-Time Communication:**

  Data is updated instantly using Firebase, ensuring no delay in approvals.

- **Complaint Management:**

  Owners can raise complaints, and admins can update their status

- **Maintenance Tracking:**

  Admin can assign maintenance charges, and owners can view dues and payment status.

- **Helper Management:**

  Owners can add house helpers and control their entry permissions.

- **Image Handling:**

  Visitor images can be managed using Cloudinary for better performance and storage.

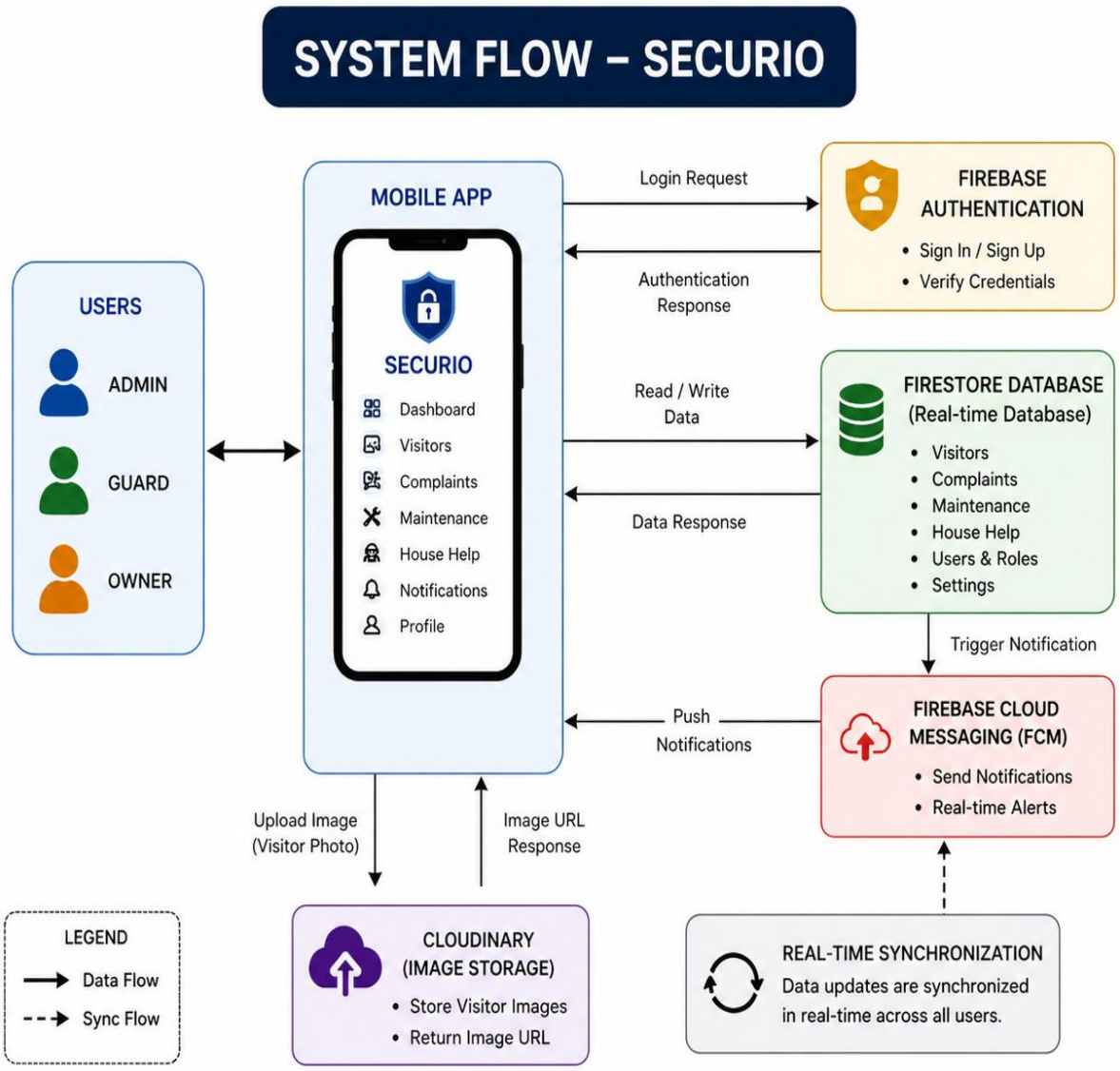

Figure 2.1: System Flow

Chapter 3: Project Outcomes

The implementation of the **Securio – Apartment Security & Management App** results in significant improvements in the overall functioning of residential societies. By replacing manual processes with a digital system, the application enhances efficiency, security, and user experience.

Key Outcomes:

- **Improved Apartment Security:**

  Digital visitor tracking ensures proper recording, verification, and monitoring of all entries and exits, reducing unauthorized access.

- **Faster Communication Between Users:**

  Real-time updates enable instant communication between guards, owners, and administrators, eliminating delays caused by phone calls.

- **Reduced Dependency on Manual Processes:**

  Automation of tasks such as visitor entry, complaint handling, and maintenance tracking minimizes paperwork and human effort.

- **Increased Transparency:**

  Complaints and maintenance records are digitally stored and updated, allowing users to track status easily and ensuring accountability.

- **Better Management of House Helpers:**

  A structured system for registering and monitoring helpers ensures controlled access and improved security.

Chapter 4: Software Requirements

The software components used in the development of the application include:

- Frontend: Android (Java/Kotlin, XML for UI design)
- Backend: Firebase Firestore (real-time database)
- Authentication: Firebase Authentication
- Notifications: Firebase Cloud Messaging (FCM)
- Image Storage: Cloudinary (for storing and optimizing images)
- IDE: Android Studio
- Operating System: Windows / Linux

Chapter 5: Project Design

The system follows a **client-server architecture**, ensuring scalability and real-time performance.

Architecture Components:

- **Client:** Android Application
- **Authentication Layer:** Firebase Authentication
- **Database Layer:** Firestore
- **Notification Layer:** FCM
- **Image Storage Layer:** Cloudinary

Modules:

- Visitor Management Module
- Complaint Management Module
- Maintenance Module
- Helper Management Module

System Flow:

1. User logs into the application
2. Role-based dashboard is displayed

3. User performs actions (add visitor, complaint, etc.)

4. Data is stored in Firestore

5. Real-time updates are sent to other users

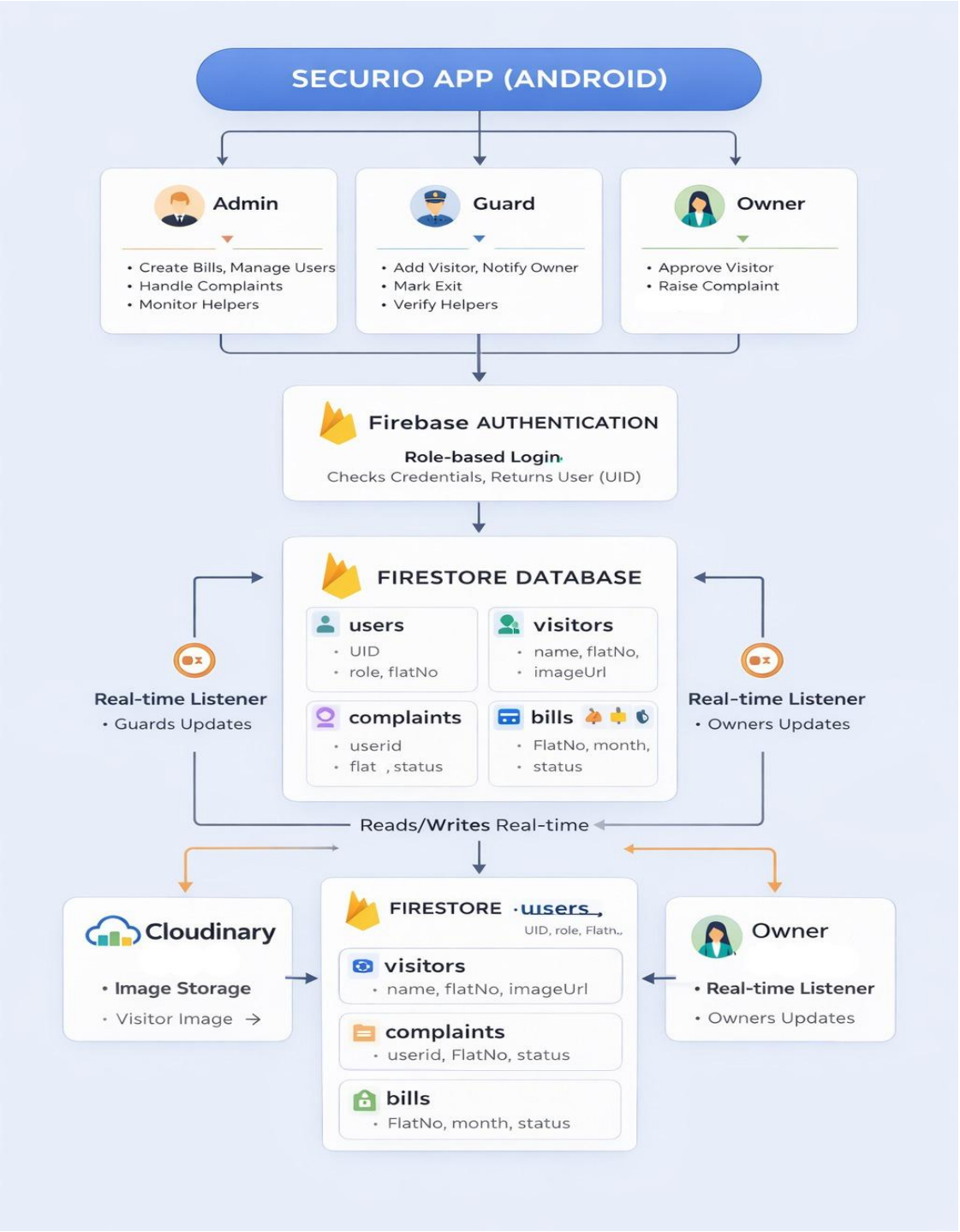

Figure 5.1: Workflow

Chapter 6: Project Scheduling

PROJECT TITLE: Securio

PROJECT GUIDE: Prof.Veena Gawde

INSTUTUTE & DEPARTMENT AP SHAH INSTITUTE OF TECHNOLOGY(CSE-Data Science)

DATE: 4-1-26

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	Project Conception and Initiation					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	Srushti Balsaraf,Shravani Bhilare	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	Sakshi Kale	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	Pratiksha ,Shravani ,Sakshi ,Srushti	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	Srushti Balsaraf	2-3-26	2-17-26	2	100%
1.5	Presentation I	Pratiksha,Shravani,Sakshi,Srushti	2-17-26	2-24-26	1	100%
2	Project Design and Implementation					
2.1	Database Design	Pratiksha Jadhav	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	Pratiksha Jadhav	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	Shravani Bhilare	3-17-26	3-24-26	1	100%
2.4	Presentation II	Pratiksha,Shravani,Sakshi,Srushti	3-24-26	4-1-26	1	100%

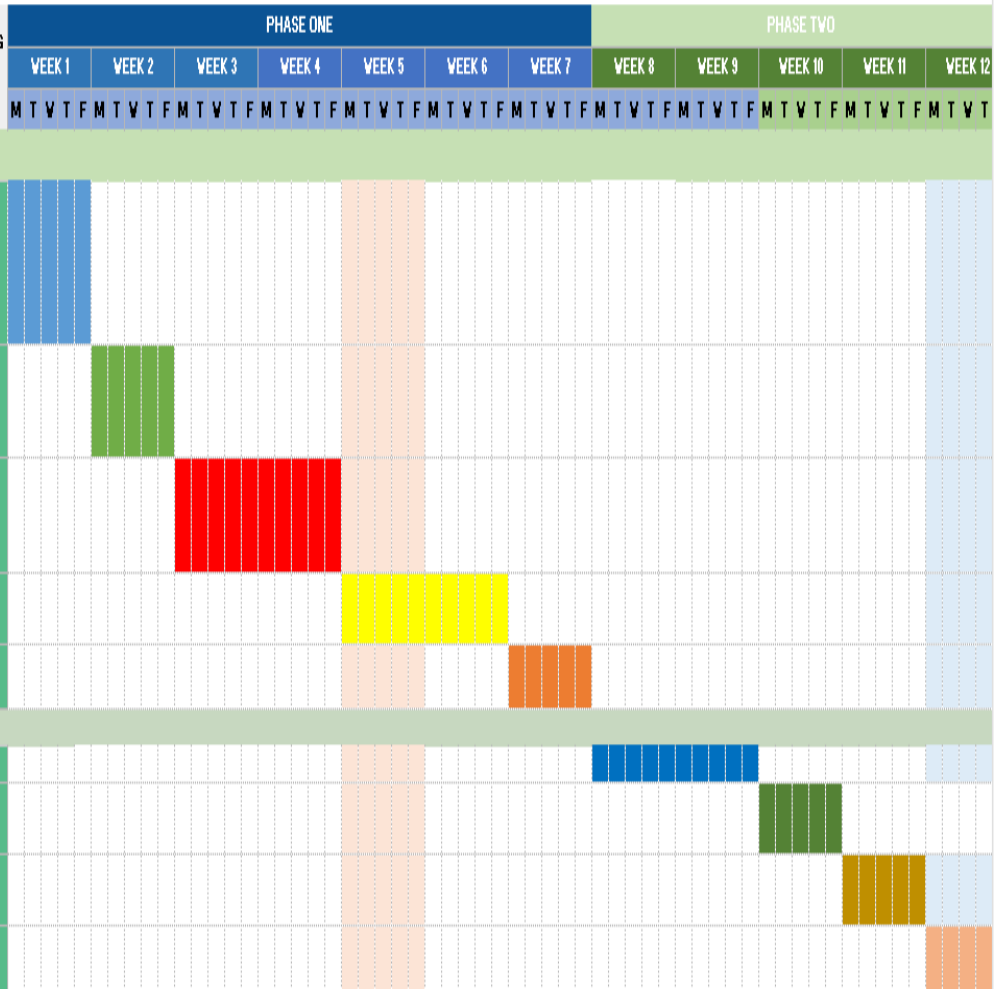

Figure 6.1: Gantt Chart

Chapter 7: Result

The **Securio – Apartment Security & Management App** was successfully developed and tested, demonstrating reliable and efficient performance across all its modules. The application includes a secure role-based login system that provides separate dashboards for Admin, Guard, and Owner, ensuring proper access control and usability.

The visitor management system operates effectively, allowing guards to add visitor details and enabling owners to approve or reject requests in real time. The complaint management module allows users to raise complaints and track their status, while the maintenance module provides accurate information about dues and payment status, ensuring transparency.

Additionally, Cloudinary integration enables efficient image handling, and the application offers a clean and responsive user interface for smooth navigation. Overall, the system performs efficiently with real-time updates, improved communication, and a user-friendly experience.

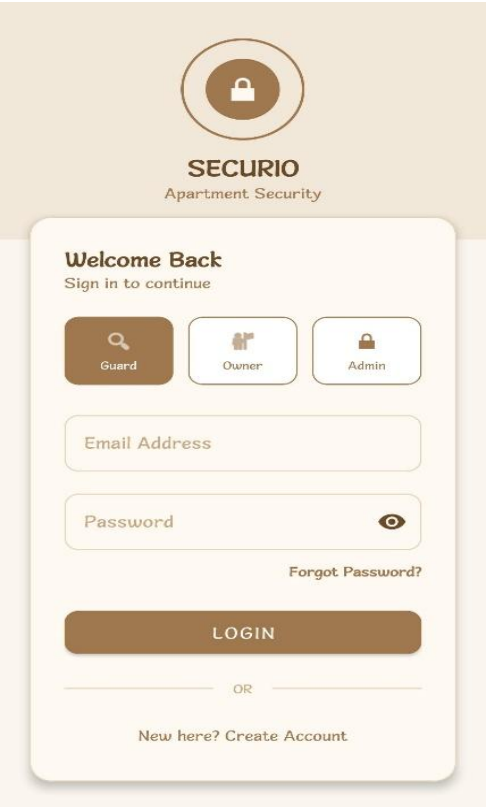

Figure 7.1: Sign-in page

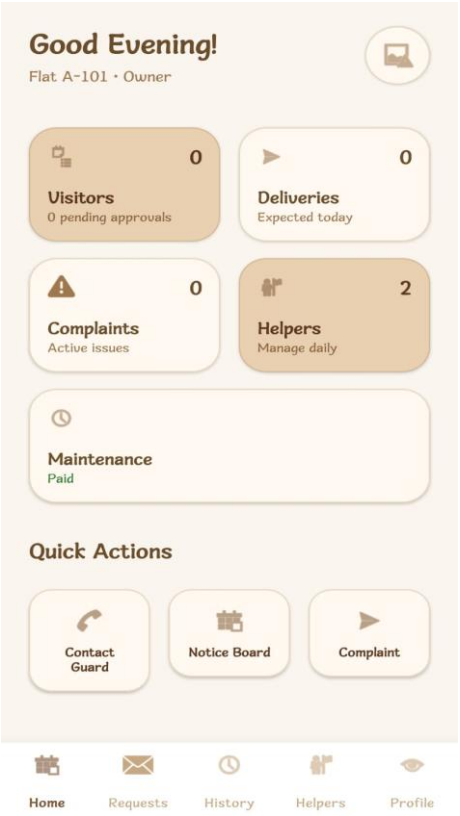

Figure 7.2: Owner dashboard

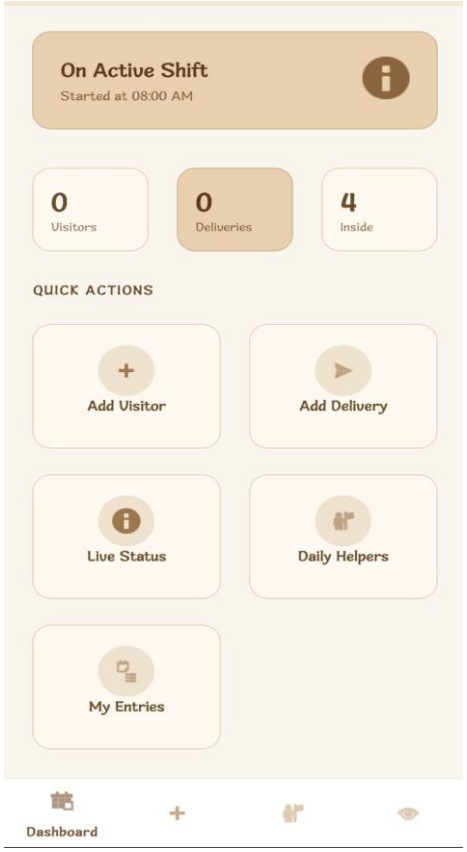

Figure 7.3: Guard dashboard

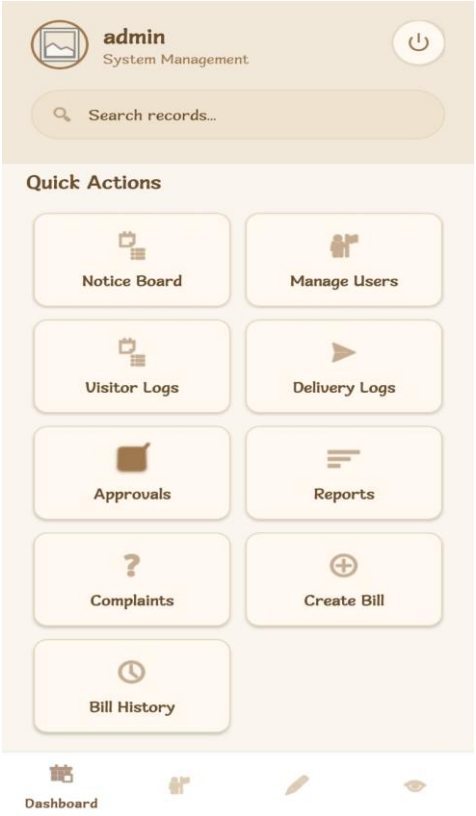

Figure 7.4: Admin dashboard

Chapter 8: Conclusion

The **Securio – Apartment Security & Management App** successfully addresses the major challenges faced by residential societies by providing a centralized and digital solution. By integrating key features such as visitor management, complaint handling, maintenance tracking, and house helper management into a single platform, the application significantly enhances security, improves communication, and simplifies overall society operations.

The use of modern technologies such as Firebase ensures real-time data synchronization, secure authentication, and efficient backend management. Additionally, Cloudinary integration enhances image handling by providing optimized storage and quick access. The role-based access system further improves usability by offering customized functionalities for Admin, Guard, and Owner, ensuring smooth coordination among all users.

The system is scalable, reliable, and user-friendly, making it highly suitable for modern residential societies. It reduces dependency on manual processes, increases transparency, and provides a seamless user experience. Overall, the application proves to be an effective solution for improving security and management in apartment complexes.

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Cloudinary Documentation (Image Storage & Optimization)

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https://www.youtube.com/

GitHub (Code Reference & Version Control)

https://github.com/

Claude AI (Idea Support & Assistance)

https://claude.ai/